

# Aamir Yousuf

Ph.D. Candidate, Computational Mathematics  
IIT Bombay and Monash University

*Curriculum Vitae*  
September 2026

## Current address

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## Research interests

Numerical methods for partial differential equations; finite element, finite difference, and hybrid high-order discretizations; coupled multiphysics and nonlinear plate problems; and stability and convergence analysis of time-dependent problems.

## Experience

- Senior Research Fellow, ANRF – J. C. Bose National Fellowship Project. March 2026–Present  
Principal Investigator: Prof. Neela Nataraj, Indian Institute of Technology Bombay (IIT Bombay).  
Project: *Numerical Analysis of Thermoelastic Diffusion in Thin and Very Thin Plates and Deep Neural-Network Solvers.*

## Education

- Ph.D. in Computational Mathematics, Joint Program of IIT Bombay (India) & Monash University (Australia) through IITB-Monash Research Academy. (Thesis submitted) 2022–Present
- Master of Science in Mathematics, University of Kashmir, Srinagar, India. 2017–2020
- Bachelor of Science, General Science, University of Kashmir, Srinagar, India. 2014–2017

## Doctoral research

**New Discretizations for Coupled Multiphysics Models in Slender Geometries** (Ph.D. thesis).

*Supervisors: Prof. Neela Nataraj (IIT Bombay) and Prof. Ricardo Ruiz-Baier (Monash University).* January 2022–Present

- Developed fully discrete schemes for coupled multiphysics and nonlinear dynamic plate problems, all modeling thin and very thin elastic plates.
- Employed lowest-order nonstandard finite elements for the spatial discretization of the fourth-order equations, conforming finite elements for the second-order equations, and finite-difference schemes for the temporal discretization of all coupled systems.
- Focused on three connected questions: stable and accurate spatial discretization, consistent and efficient discretization of coupled terms, and reliable time integration across equations with different time-stepping schemes.
- Proved well-posedness, stability, and optimal-order spatial and quadratic temporal convergence for all schemes, and designed energy-conserving, unconditionally stable schemes for nonlinear plate problems.

## Academic projects

- Fully discrete scheme for the biharmonic wave equation.** July–October 2023  
A course project for MA 855 (Topics in Numerical Analysis I) with instructor: Prof. Neela Nataraj, IIT Bombay.  
Developed a fully discrete numerical scheme for the biharmonic wave equation and carried out its analysis.
- Finite element discretization of the Euler–Bernoulli beam theory.** January–May 2022  
A course project for CE 620 (Finite Element Methods) with instructor: Prof. Mandar Inamdar, IIT Bombay.  
Developed the finite element discretization for the Euler–Bernoulli beam theory, jointly with two other project members.

## Publications

### Journal articles

- N. Nataraj, R. Ruiz-Baier, A. Yousuf, “Unified numerical analysis for thermoelastic diffusion and thermo-poroelasticity of thin plates,” *Mathematical Models and Methods in Applied Sciences*, 2026. [PDF](#)
- N. Nataraj, R. Ruiz-Baier, A. Yousuf, “Semi- and fully-discrete analysis of lowest-order nonstandard finite element methods for the biharmonic wave problem,” *Computational Methods in Applied Mathematics*, 25(4), 921–948, 2025. [PDF](#)

### Preprints

- C. Carstensen, N. Nataraj, R. Ruiz-Baier, A. Yousuf, “Unconditionally stable and energy-conserving discretization of the dynamic von Kármán equations,” *arXiv:2608.14087*, 2026. [PDF](#)

4. R. Kumar, N. Nataraj, A. Yousuf, “Hybrid-high order method in space and implicit schemes in time for the biharmonic wave equation,” *arXiv:2606.15163*, 2026. [PDF](#)

## Honors and awards

### Fellowships and scholarships

- Senior Research Fellow, ANRF – J. C. Bose National Fellowship Project, IIT Bombay. March 2026–Present
- IITB-Monash Research Scholar, Collaborative Ph.D. Program. January 2022–Present
- DBT Fellow, Department of Biotechnology, Government of India. January 2022–Present
- Qualified CSIR-NET with All India Rank 33. 2022
- Qualified GATE with All India Rank 157 and All India Rank 364. 2021 & 2022

### Awards and recognition

- Student Member, Australian Mathematical Society. January 2022–Present
- Conference Program Compiler, CTAC 2024, Monash University, Melbourne, Australia. 2024
- Sports Person of the Year, Hostel-13, IIT Bombay. 2025
- Sports Citation of the Year, Hostel-13, IIT Bombay (highest inter-hostel sports honor). 2023
- Member, Inter-IIT Cricket Team, IIT Bombay; participated in national-level tournaments. 2022–2025

## Teaching experience (Teaching assistant)

| Course                                     | Institution       | Period                |
|--------------------------------------------|-------------------|-----------------------|
| SI 507 – Numerical Analysis                | IIT Bombay        | July–October 2025     |
| MTH1010 – Functions and their Applications | Monash University | March–June 2025       |
| MA 214 – Numerical Analysis                | IIT Bombay        | January–May 2024      |
| MA 111 – Calculus II                       | IIT Bombay        | July–October 2023     |
| SI 416 – Optimization                      | IIT Bombay        | February–May 2023     |
| MA 105 – Calculus I                        | IIT Bombay        | January–February 2023 |

## Conferences and invited talks

- Scientific Computing and Differential Equations (SciCADE-26), Edinburgh, Scotland, UK.  
*Funded by the Institution of Eminence (IoE), IIT Bombay.* 29 June–3 July 2026
- Advanced Computational Methods in Engineering and Applied Mathematics (ACOMEN-25), Ghent, Belgium.  
*Funded by IIT Bombay.* 15–19 September 2025
- Australia and New Zealand Industrial and Applied Mathematics (ANZIAM-25), Coffs Harbour, Australia.  
*Funded by IITB-Monash student funding, Monash University.* 02–05 February 2025
- Joint Meeting of the NZMS, AustMS and AMS 2024, University of Auckland, New Zealand.  
*Jointly funded by Monash University and AustMS Student Support Scheme.* 09–13 December 2024
- Computational Techniques and Applications Conference (CTAC-24), Monash University, Melbourne, Australia.  
*Funded by the School of Mathematics, Monash University.* 19–22 November 2024
- Latest Advances in Computational and Applied Mathematics (LACAM-22), IISER Thiruvananthapuram, India.  
*Funded by IITB-Monash Research Academy.* 21–24 February 2022

## Schools and short courses

- Discrete de Rham Method, Prof. Jérôme Droniou (University of Montpellier).  
*Venue: IIT Bombay, India.* February 2026
- The Mathematics of Liquid Crystals, Prof. Apala Majumdar (University of Manchester).  
*Venue: IIT Bombay, India.* February 2026
- Optimality of Adaptive Finite Element Methods, C. Carstensen, L. Diening, C. Kreuzer, D. Praetorius.  
*Venue: Bonn University, Germany.* September 2025
- Adaptive Finite Element Methods, Prof. Carsten Carstensen (Humboldt University Berlin).  
*Venue: Monash University, Australia.* December 2024
- Introduction to HHO Methods, Prof. Jérôme Droniou and Prof. Daniele Di Pietro (University of Montpellier).  
*Mode: Online.* April 2024

## Technical skills

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- **Programming languages:** C++, Python.
- **Numerical software:** FEniCS, FreeFEM, MATLAB, Firedrake.
- **Tools:** L<sup>A</sup>T<sub>E</sub>X, Excel, MS Office.

## Additional information

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- **Strengths:** Mentoring, adaptability, inquisitiveness.
- **Languages known:** English, Urdu, Kashmiri, Hindi, Arabic (beginner).
- **Interests:** Playing and watching cricket, traveling, trekking.

## Ph.D. Coursework

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| Code    | Course Title                                            | Credits |
|---------|---------------------------------------------------------|---------|
| MA 855  | Topics in Numerical Analysis I                          | 6       |
| MA 854  | Topics in Differential Equations II                     | 6       |
| MA 824  | Functional Analysis                                     | 6       |
| MA 817  | Partial Differential Equations I                        | 6       |
| MA 814  | Complex Analysis                                        | 6       |
| MA 813  | Measure Theory                                          | 6       |
| AE 738  | Tensors for Engineers                                   | 6       |
| CE 620  | Finite Element Methods                                  | 6       |
| MA 540  | Numerical Methods for Partial Differential Equations    | 6       |
| MA 503  | Functional Analysis                                     | 8       |
| MES 801 | Seminar                                                 | 4       |
| ME 899  | Communication Skills                                    | 6       |
| TA 101  | Teaching Assistant Skill Enhancement & Training (TASET) | –       |
| GC 101  | Gender in the Workplace                                 | –       |

## References

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### 1. Prof. Neela Nataraj

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### 2. Prof. Ricardo Ruiz-Baier

Professor of Computational Mathematics  
School of Mathematics, Monash University  
Office 315, 9 Rainforest Walk  
Melbourne, VIC 3800, Australia

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### 3. Prof. Harsha Hutridurga

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